Low-Frequency RoPE Subspace for Near–Far Sparse Attention

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***Abstract*—Page-to-token sparse attention faces two coupled decisions: how to score a long history cheaply, and how to allocate a limited candidate set before final attention. We study a near–far cascade that uses the same 32 low-frequency RoPE coordinates for page screening and token refinement, reserves page quotas for both regions, and recomputes final attention with original-dimensional queries, keys and values. Qwen3-8B diagnostics show why these decisions interact: regional reservation raises mean far-region attention retention from .6759 to .7207, while reducing it in 43 of 128 correlated observations. The budget analysis further shows that reserving far pages alone need not leave any far-token quota. Historical five-task LongBench scores are 47.588 for shared-subspace scoring and 47.910/47.666 for nominal full-dimensional page/token variants, but do not isolate representation effects. Separate timing studies find no selector-time advantage over nominal full-dimensional page scoring and higher sparse decode time than dense attention. Thus lower scoring dimensionality and regional access are insufficient to establish an efficient sparse decoder. The compact implementation with 4/128 prefix/recent protection still requires matched quality and preparation-inclusive cost evaluation.**

***Index Terms*—sparse attention, shared representation, regional retrieval, long-context inference**

I. Introduction

Long-context decoding repeatedly accesses previously generated keys and values. Sparse attention limits this work by selecting a smaller token set, but selection introduces its own scoring, ranking and memory accesses. A page-to-token cascade first decides which pages may contribute tokens to refinement. The aim is to retain useful candidates within a fixed token budget at a retrieval cost below the attention work saved.

Compact page summaries reduce scoring cost but can obscure rare high-scoring tokens. Averaging can dilute their responses; coordinate-wise extrema from different tokens can yield loose upper bounds, admitting pages with no correspondingly high-scoring token. Following ClusterKV's token/page perspective [7], Figure 1(a) shows that both mean and extrema selections miss pages in the same page-mass ranking reference. With enough positive-probability candidates to fill the quota, refinement cannot recover unprotected tokens from rejected pages. Summaries must therefore be assessed by candidate access as well as scoring cost.

A global TopK budget lets high-scoring candidates from one region occupy slots without guaranteeing minimum capacity for the other. In Figure 1(b), separate near and far reservations raise mean far retention from .6759 to .7207 while lowering it in 43 of 128 correlated observations. This controllable coverage trade-off calls for explicit page and token capacities: reserving far pages alone need not preserve far tokens.

These observations lead to a cascade with two explicit controls. A fixed set of complete low-frequency RoPE pairs supplies the scoring coordinates for both stages; separate near–far page quotas determine regional access to token refinement. The coordinate diagnostic in Figure 1(c) motivates the compact view, although its unpaired random baseline does not isolate frequency from pair structure. Original-dimensional Q/K/V are retained for final attention, so approximation enters through token selection. We also consider an optional prefill-informed far gate. The historical layer variation in panel (d) motivates this extension, but uses a different single-query statistic; the gate is disabled by default and must be assessed independently.

We formulate shared-coordinate regional selection and analyze how page and token budgets jointly constrain retrieval. The evaluation examines task quality, regional coverage and execution cost. Qwen3 diagnostics show changes in coverage, but historical scores do not isolate either mechanism and sparse decoding is slower than dense attention. The compact 4/128-protection implementation still lacks matched quality–cost evaluation.

II. Related Work

Quest, SparQ and HISA use query-aware approximate selection to reduce the active attention set [1], [2], [3]. We adopt Quest's sign-aware page extrema for far-page scoring. AsyncTLS already combines a page-to-token cascade with coordinate selection, quantization, per-query normalization followed by GQA probability aggregation, and final attention over original KV [4]. We study these existing components with complete low-frequency RoPE pairs shared across both stages and independent near–far page quotas. Our L2 path scores the full history before masking.

Reducing attention traffic also requires considering what the selector itself reads. Token-Picker estimates probabilities from partial key bits and schedules further accesses on demand [12]; our coordinate-based scorer instead retains full-dimensional KV and derives auxiliary scoring views. KV compression and placement methods address complementary storage and movement constraints [5], [6], [7]. A smaller scoring view in our design reduces neither original-KV capacity nor, by itself, complete-request latency.

III. Near–Far Selection in a Shared Subspace

To reduce candidate-scoring width, the cascade uses one 32-coordinate map for L1 page screening and L2 token refinement (Fig. 2). L1 selects near and far pages under separate quotas. L2 scores the full history, masks by retained pages, and selects regional tokens. Their indices merge with protected positions for original-dimensional Q/K/V attention; recomputed logits do not remove support-selection error.

A. Shared low-frequency coordinates

For Qwen3-8B, we select post-RoPE coordinates $J = \{48, \dots, 63, 112, \dots, 127\}$ from each 128-dimensional query and key. These are 16 complete low-frequency split-half RoPE pairs, not a contiguous suffix or learned projection (Fig. 2a). With coordinate-selection matrix $E_J \in \mathbb{R}^{128\times 32}$, the scoring vectors are $z_q = E_J^\top q$ and $z_i = E_J^\top k_i$. The same map serves both regions and stages within each layer and KV head, while query heads retain their own scores. Slower positional phase variation motivates this choice; it does not guarantee retrieval fidelity. The compact implementation uses width-32 scoring tensors, whereas the historical quality runs used 32 active coordinates in width-128 L2 tensors (S7).

L1 derives page summaries and L2 derives quantize–dequantize token keys from the same coordinate map. The L2 keys use 16 floating-point quantize–dequantize levels, not packed four-bit storage or INT4 arithmetic (S7). L1 summaries and L2 keys occupy separate buffers; both regions use the same coordinates. Original keys and values remain full width. These auxiliary views add preparation and maintenance cost without compressing the original KV cache (S7).

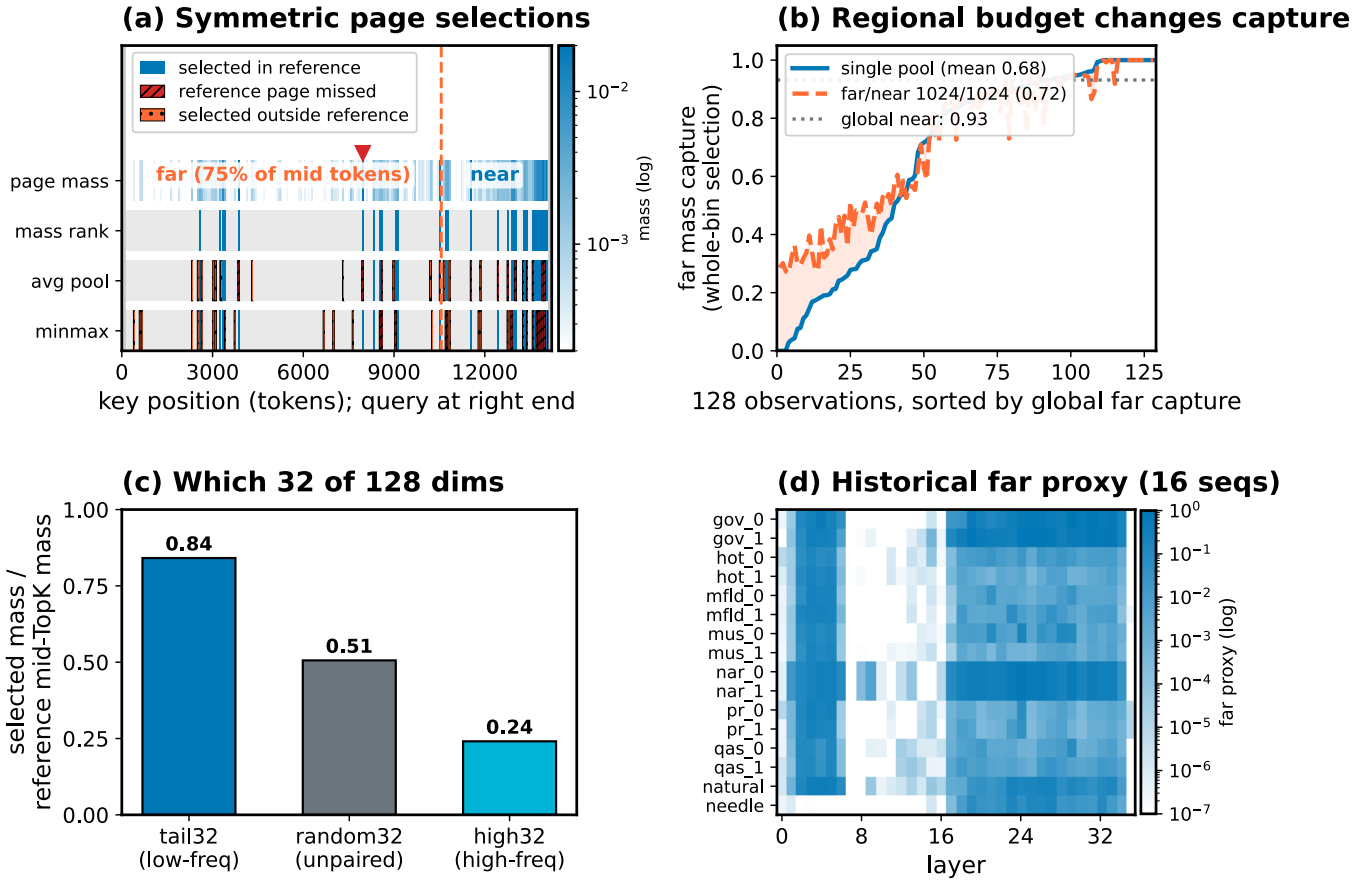


Fig. 1. Four motivating diagnostics; Qwen3-8B, 128/128 prefix/recent protection in (a)–(c). (a) MuSiQue sample 1, layer 24, query 14212. The top strip shows summed attention per page, not a selection. Below are saved mass-rank, avg and minmax sets: 32 pages each, using 1989/1989/2048 tokens under cap2048. Blue denotes selected reference pages; red hatching denotes missed reference pages; orange dots denote selected pages outside the reference. Mass-rank is a whole-page comparison reference, not an optimal oracle, exact token TopK or DSA. Representation and pooling choices differ between these saved selections; their misses do not isolate a pooling effect or establish a quality ranking. (b) Four sequences contribute 128 correlated observations, ordered by global far retention. A global whole-bin cap of 2048 is compared with separate far/near caps of 1024/1024, without refill; red shading marks lower capture. Retention divides selected regional mass by full regional mass; the dotted line is global near retention. (c) Selected mass relative to full-dimensional reference mid-TopK mass; random32 does not preserve RoPE pairs. (d) The same 16-by-36 historical single-query proxy matrix gives 24/36 layer medians below .01 and 12–32 low-proxy layers per sequence. It uses the last sampled query, group-logit summation before softmax and far interval $[64, t-2048)$ (S6), unlike the current gate. These are diagnostic selections and profiles, not task-quality or decode-safety results.

B. Regional page and token budgets

In decoding, zero-based position t includes prefill, and $S = t+1$ counts valid cached positions through the current token, excluding padding. Let s and w denote the protected prefix and recent-window lengths. Their union and the unprotected middle are

$$\mathcal{P}_t = \{i : 0 \le i \le t,\ i < s \text{ or } i > t-w\}, \tag{1}$$

$$\mathcal{M}_t = \{0, \dots, t\} \setminus \mathcal{P}_t. \tag{2}$$

Overlapping protection counts once. Near is a suffix of $\mathcal{M}_t$; far is its complement there. Near excludes the protected recent window. Near uses fraction α of unprotected history, a one-page minimum and a page-aligned start; S7 specifies integer rules.

Regional page caps and token quotas constrain the cascade (Fig. 2b). Near uses query–average-key dot products; far uses sign-aware extrema. This regional assignment is a heuristic. Summaries cover full physical pages, including protected boundary positions; L2 separately limits token regions (S7). For page b, let $m_{b,j} = \min_{i \in b} z_{i,j}$ and $M_{b,j} = \max_{i \in b} z_{i,j}$. Quest's construction [1] gives

$$U_b(z_q) = \sum_j (z^+_{q,j} M_{b,j} + z^-_{q,j} m_{b,j}), \tag{3}$$

where $z^+ = \max(z, 0)$ and $z^- = \min(z, 0)$. Each signed coordinate contribution is bounded by its chosen extremum, so $z_q^\top z_i \le U_b(z_q)$ for every unquantized shared-space key in b. This does not bound original-dimensional logits or independently quantized scores. Extrema may come from different tokens, making the bound loose enough to misrank pages and miss important tokens. Average-key scores lack this per-token bound.

For total cap $K \ge |\mathcal{P}_t|$, define the middle-token budget $K_{\text{mid}} = K - |\mathcal{P}_t|$. α controls near extent, β its share of L1 page cap K_1, and γ its nominal L2 quota from retained-page capacity. For the partitioned fixed-quota path (far active; no overrides), assume integer $K_1 \ge 1$, $\alpha > 0$, $0 < \beta \le 1$ and $\gamma \ge 0$. With page size B and sufficient regional pages/tokens, $P_n = \max(1, \text{round}(\beta K_1))$, $P_f = \max(0, K_1 - P_n)$, $k_n = \min(\lfloor \gamma P_n B \rfloor, K_{\text{mid}})$ and $k_f = \max(0, K_{\text{mid}} - k_n)$; rounding uses ties-to-even. Thus $P_f > 0$ does not imply $k_f > 0$: page and

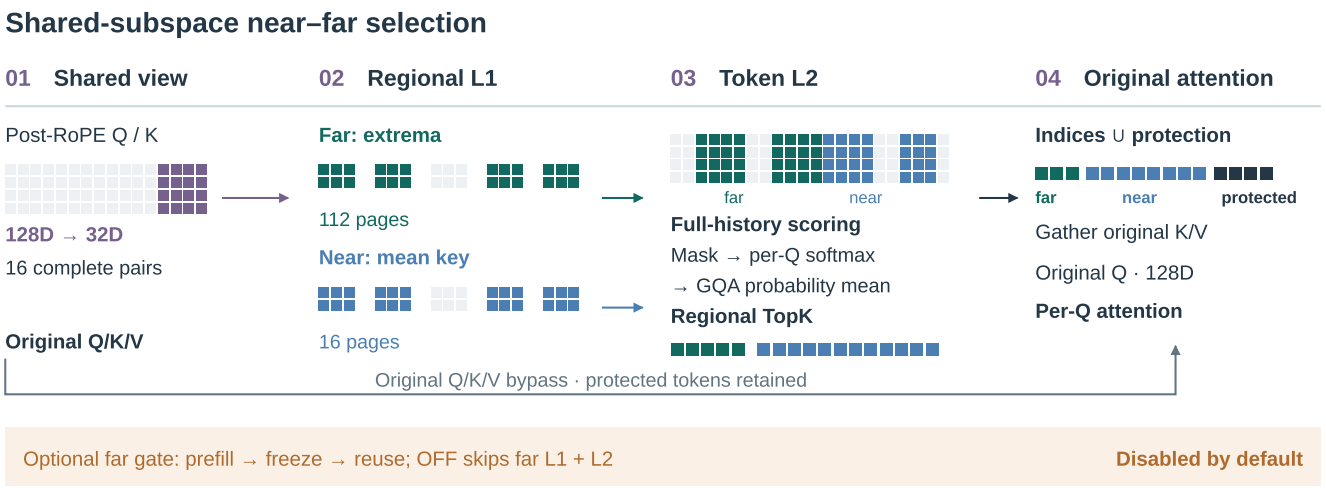


Fig. 2. Shared-subspace selection and the original-Q/K/V bypass. L1 reserves regional pages; L2 masks full-history scores, normalizes each query over shared retained support, averages probabilities within a KV group, and selects regional tokens. When OFF, the optional gate skips far scoring/selection and refills near L2 within retained near pages. Counts illustrate the sufficient-capacity configuration.

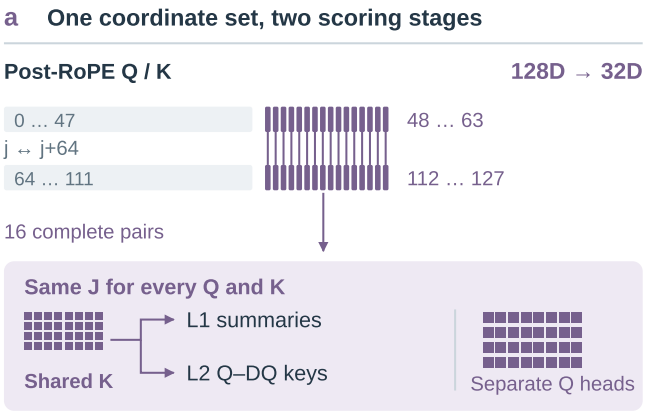


Fig. 2a. Sixteen complete split-half RoPE pairs form the 32-coordinate scoring view. Per layer and KV head, shared coordinates supply separate L1 summaries and quantize–dequantize L2 keys; query views stay separate. Original Q/K/V bypass these derived views (Fig. 2).

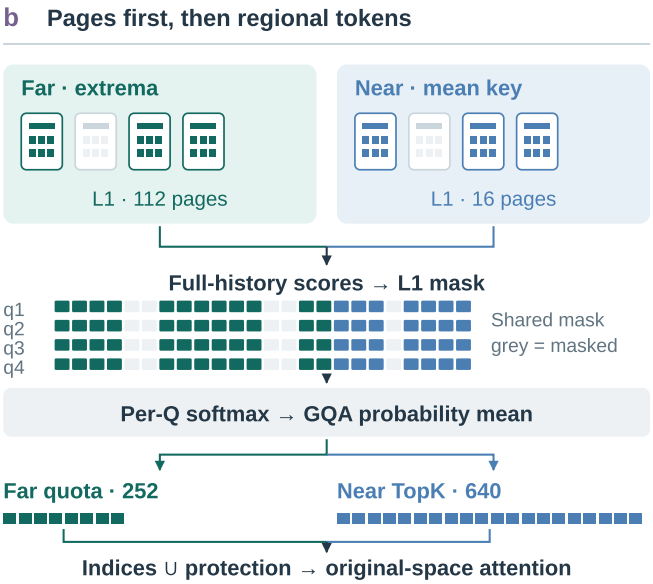


Fig. 2b. Regional screening with full-history L2 scoring. Colored cells indicate eligible candidates, not measured scores. With disjoint protection and sufficient capacity, compact 4/128 uses 112/16 far/near pages, nominal 252/640 slots and 132 protected tokens under $K = 1024$. Subpage prefix reselection can consume up to four far slots. Near excludes the protected recent window.

token access are separate constraints. For $K_1 = 128$ and $\beta = .125$, L1 reserves 16 near and 112 far pages. With enough positive-probability candidates, L2 cannot recover unprotected tokens from L1-rejected pages. S7 details scarce-support and masked zero-tie exceptions.

With disjoint protection and sufficient candidates, the compact configuration ($K = 1024$, $s = 4$, $w = 128$) leaves 892 middle slots per KV head. Here $\gamma = .625$ yields $16 \cdot 64 \cdot .625 = 640$ near and 252 far slots (Supplementary Fig. S2). The partial prefix page can reselect up to four protected tokens, so 252 is nominal rather than guaranteed unprotected far coverage.

C. GQA selection and original-dimensional attention

Grouped-query attention (GQA) assigns four of Qwen3's 32 query heads to each of its eight KV heads. L1 averages their page scores before regional TopK. Ordinary L2 computes each query's full-history logits, applies the shared page mask, then softmax. Group-averaged probabilities determine regional token TopK. L2 uses scale $1/\sqrt{128}$ before softmax even with 32 scoring coordinates. Because softmax precedes group averaging, this rule generally differs from applying softmax to group-averaged logits. Width-32 scoring narrows each dot product; a smaller L1 shortlist restricts eligibility but does not reduce ordinary L2's full-history dot-product count.

For KV group h, let $\mathcal{S}_{t,h}$ be the deduplicated union of protected positions and regional winners, with $|\mathcal{S}_{t,h}| \le K$. Each query g uses this support for its own full-dimensional attention ($d = 128$):

$$o_{h,g} = \sum_{i \in \mathcal{S}_{t,h}} \frac{e^{q_{h,g}^\top k_{i,h}/\sqrt{d}}}{\sum_{j \in \mathcal{S}_{t,h}} e^{q_{h,g}^\top k_{j,h}/\sqrt{d}}} v_{i,h}. \tag{4}$$

One softmax over merged indices avoids adding separately normalized

regional outputs [8]. Full-dimensional retained logits still use a restricted denominator, which can change the output. Physical KV gathering and reuse depend on backend and length; short sequences can use a dense fallback.

D. Prefill-informed far gating

The optional, default-disabled gate uses prefill observations to decide whether to omit far selection. An input–layer mean far-attention weight sum below ϵ sets OFF; otherwise selection stays active. The first decode-path call freezes the decision until input reset (S5). OFF skips far L1/L2 scoring and selection in the supported fixed-quota path, refilling near L2 within unchanged near pages. Removing far support can change softmax denominators and group-averaged near rankings (S5). Representation preparation and sequence-wide score buffers and normalization remain.

The gate supports one sequence; the skip path requires valid length and nonempty far history (Fig. 2c; S5). The prefill threshold does not bound decoding error; gate quality and total cost require separate evaluation.

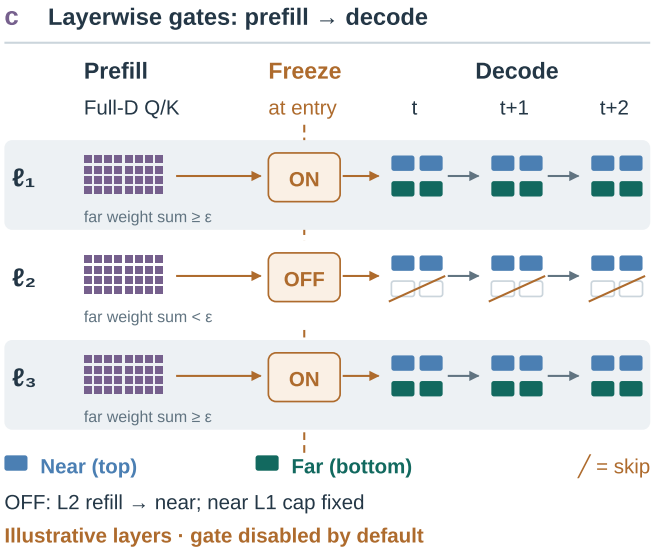


Fig. 2c. Optional far-only control, disabled by default. Full-dimensional causal prefill far-attention weight sum, averaged over the last ≤ 256 available query rows and all query heads, determines each input–layer decision. The first decode-path call freezes it until input reset. OFF skips far L1/L2 and refills near L2 within unchanged near pages. Layer states are illustrative; paired cells denote L1/L2. This statistic does not bound decoding error.

IV. Experimental Setting

We evaluate three aspects separately: task quality, regional candidate coverage and execution cost. Table I defines the compact Qwen3-8B design with dense prefill, sparse decoding and a 1024-token cap including four prefix and 128 recent tokens. The available LongBench measurements instead use an earlier shared-subspace implementation: 128/128 protection and 32 active coordinates stored in zero-filled width-128 L2 keys. The comparison labels "full-dimensional page scoring" and "full-dimensional token scoring" denote reported variants whose complete execution configurations are unavailable. We retain these nominal labels without treating the runs as controlled single-stage ablations. Synthetic selector timing and model-level decode timing come from separate studies; none supplies matched quality and cost for the compact design. S6/S7 give configuration and measurement details.

TABLE I
Compact design parameters and execution policy. These settings define the proposed implementation, not the historical runs. K includes unique protected tokens.

Parameter	Setting
Final token cap K	1024
Page size; L1 page cap K_1	64; 128
α, β, γ	.25, .125, .625
L1/L2 active coordinates	32/32
Prefix / recent protection	4/128
Prefill; decode layers	dense; all sparse
L2 scoring support	full history before mask

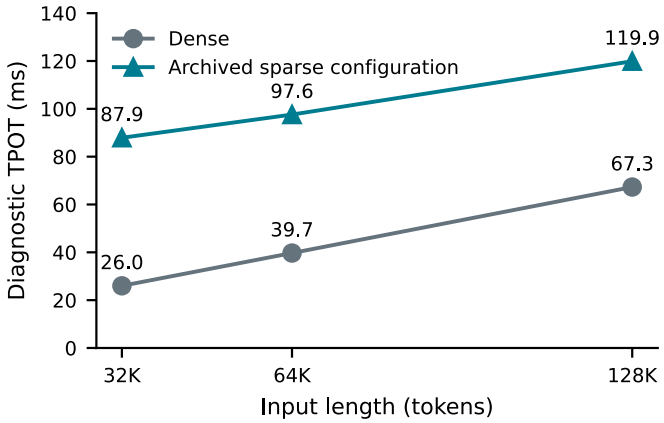


Fig. 3. Author-reported model-level decode TPOT for an archived sparse configuration and dense FullKV, Qwen3-8B with dynamic cache. Fixed 128-token output gives 127 after-first intervals, including device synchronization and argmax. The archived $(.125, .375, .625)$ setting differs from the selected 4/128 candidate. These medians do not measure complete-request E2E or establish a matched quality–cost comparison; provenance limits are in S6.

Workloads and comparisons. The quality evidence comprises five LongBench tasks and a historical 11-task RULER [10] 128K comparison. The LongBench scores compare three nominal sparse scoring configurations; dense FullKV is a separate reference in the RULER and decode-timing records. Controlled sparse comparisons must align final unique-token cap K (including protection), model, inputs and phase policy. Realized token counts and head-sharing granularity also affect KV traffic; internal page budgets need not match (S7).

Metrics. Quality is the unweighted mean of task scores; task-specific changes are reported where available. Candidate retention measures selected attention mass divided by the corresponding full regional mass. Selector time covers a synthetic single-layer operation, whereas synchronized TPOT measures model-level intervals after the first output token. Neither is complete-request latency, which also includes preparation, maintenance and input/output work. Because the quality and timing studies are unpaired, they cannot define a quality–latency frontier. Five task means also do not establish document-paired non-inferiority; that requires matched predictions and a prespecified acceptable-loss margin.

V. Evaluation and Discussion

A. Q1: Does reduced scoring width preserve task quality?

The five-task means on HotpotQA, Qasper, MuSiQue, GovReport and RepoBench-P [9] are 47.588 for shared-subspace scoring, 47.910

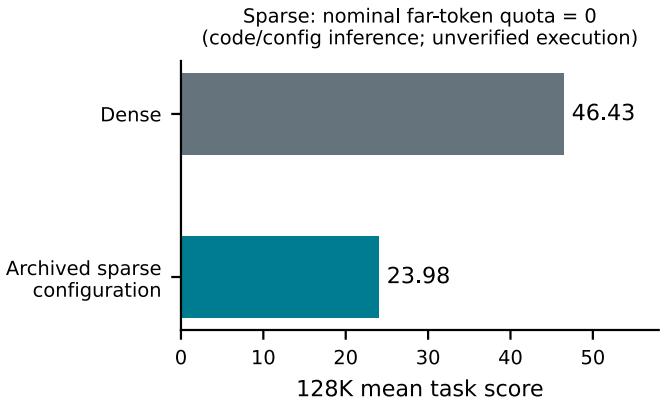


Fig. 4. Author-reported RULER 128K scores: 11 tasks, 100 examples per task and arm. The archived sparse configuration scores 22.45 points below dense FullKV; their macro-average task-score ratio is 51.6%. Reported $\beta = .375$, $K_1 = 128$ and 128/128 protection imply a nominal far-token quota of zero under the inspected formula (middle budget 768; S6). This is a code/configuration inference, not verified execution or causal attribution. The historical setting does not validate the selected 4/128 candidate or support sample-paired inference.

for the nominal full-dimensional page variant and 47.666 for the nominal full-dimensional token variant (Supplementary Fig. S4). The shared-subspace score is lower by 0.322 and 0.078 points, respectively. These differences describe the recorded configurations; they do not show that reducing scoring width preserves quality. The variants are not verified single-factor interventions, and the shared-subspace run uses the earlier 128/128-protection implementation. Consequently, neither stage's representation effect nor the compact 4/128 design's quality can be inferred from this comparison. All three configurations are sparse; dense FullKV is a different reference.

B. Q2: How do regional budgets constrain candidate access?

Regional reservation changes which distant tokens remain reachable. In Figure 1(b), separate quotas increase mean far retention but decrease it for 43/128 observations, illustrating the cost of redistributing a fixed budget. This diagnostic uses whole-bin selection on four sequences, so it does not measure production-selector task quality. The two-stage budget equations identify a second constraint: even a retained far page contributes no nominal far-token slots when the near quota exhausts K_{mid}. Substituting the historical RULER setting ($\beta = .375$, $K_1 = 128$, $B = 64$, $\gamma = .625$, protection 128/128) gives $K_{\text{mid}} = 768$ and $k_f = 0$. Figure 4 reports scores of 23.98 versus 46.43 for dense attention, a 22.45-point loss, but the algebra alone neither verifies its executed path nor attributes the loss to this quota. The current compact candidate lacks a matched dense comparison.

The protection-only configuration scores 26.09 against a 47.91 reference, while using 256 rather than 1024 tokens. The difference therefore combines retrieval-policy and budget effects.

C. Q3: Does compact scoring reduce execution cost?

Model-level decoding. The reported sparse configuration is slower than dense FullKV at every measured length (Fig. 3). Sparse TPOT medians are 87.9, 97.6 and 119.9 ms at 32K, 64K and 128K, compared with dense medians of 26.0, 39.7 and 67.3 ms. The corresponding ratios are 3.38, 2.46 and 1.78. Although the relative penalty decreases with length, absolute sparse TPOT increases. Thus these measurements show no decode-time benefit from that implementation. They include device synchronization and argmax, exclude tokenization, input transfer and text output, and use a configuration different from the compact 4/128 design. Aggregate medians cannot locate the cause of the slowdown or predict the compact implementation's performance; repetition and execution-path details are in S6.

Local selector cost. A separate two-round synthetic experiment (single layer, batch one, one query, 64K QKV) reports steady-state times of 1.3502 ms for nominal full-dimensional page scoring, 1.3598 ms for shared-subspace scoring and 1.4634 ms for nominal full-dimensional token scoring. The shared-subspace selector thus has no lower reported time than the nominal page-scoring variant in this setting. An unresolved sampling-protocol discrepancy limits interpretation of small differences (S6). These timings neither locate a bottleneck nor pair with LongBench quality or complete-request cost.

D. Optional gate: prefill is not a decode guarantee

Prefill statistics also fail to bound later far-region demand: an observed input–layer pair with prefill far-attention weight sum below .05 reaches .1678 during decoding. Relative to the reported reference score of 47.910, gate thresholds of .05 and .10 are associated with mean decreases of .034 and .218 points, respectively; three of five tasks decline in each configuration, including HotpotQA decreases of 1.05 and 1.26 points. The larger per-task losses demonstrate that a small average change can conceal task sensitivity.

The gate summaries omit the complete scoring configuration, precluding attribution to gated shared-subspace selection (S6).

Design implications and limitations. Three constraints emerge. First, scoring width controls the representation used for ranking, but the current L2 path still performs full-history scoring and normalization. Second, regional page access and regional token access require separate budgets; satisfying the first does not ensure the second. Third, the cost of preparing and using a selector must be recovered by reduced attention work. The reported timings do not show that recovery. The missing matched compact evaluation prevents a quality–cost conclusion, and the historical gate summaries do not identify the shared-subspace configuration. The evidence therefore supports the mechanism and these constraints, while leaving the proposed combination's comparative benefit unresolved.

VI. Conclusion

We studied sparse attention as a joint representation and allocation problem. A shared low-frequency coordinate map supplies page and token scoring, regional quotas control access to refinement, and final attention uses original-dimensional Q/K/V. The analysis distinguishes low-dimensional ranking from reduced execution work, and far-page reservation from final far-token coverage. Qwen3 diagnostics illustrate these trade-offs, including regional-retention losses, substantial historical quality loss and slower sparse decoding. Establishing a useful operating point for the compact 4/128 design requires measuring quality and preparation-inclusive cost under the same configuration; the available separate studies do not yet establish that benefit.

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